

Load Interconnection System Impact Study Report

For Queue Project Q12-0115

Chincoteague 138 kV

January 25, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q12-0115. The Interconnection Customer proposes a 171.0 MW large load (data center) facility interconnecting at the Chincoteague 138 kV substation in the Virginia area. The request was received on March 11, 2012. The study evaluated system intact and single contingency conditions using the regional planning case, including 80 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 20 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$81,006,000.

3. Network Impacts

3.1 Load Deliverability Analysis

1. Facility: WATTSVILLE – CHINCOTEAGUE 138 KV
Contingency: TASLEY 345/138kV Autotransformer
Loading: 142.5% (Rating: 199 MVA)
DFAX: 100.0%
Pre-project loading: 56.5%
2. Facility: W -STATION – ST BONIFACIUS 138 KV Line
Contingency: VIENNA OPERATIONS – SOUTHTOWN 345 KV
Loading: 138.0% (Rating: 263 MVA)
DFAX: 54.3%
Pre-project loading: 102.7%
3. Facility: SHINGLE CRK SUBSTATION – W -STATION 138 KV L
Contingency: SOUTHTOWN – VIENNA OPERATIONS 345 KV
Loading: 136.0% (Rating: 258 MVA)
DFAX: 54.3%
Pre-project loading: 99.9%
4. Facility: SUNSET – ST BONIFACIUS 138 KV
Contingency: SOUTHTOWN – VIENNA OPERATIONS 345 KV L
Loading: 133.5% (Rating: 288 MVA)
DFAX: 54.3%
Pre-project loading: 101.2%
5. Facility: OCEAN CITY – REIDLAND 138 KV line
Contingency: SOUTHTOWN – VIENNA OPERATIONS 345 KV
Loading: 129.7% (Rating: 190 MVA)
DFAX: 45.6%
Pre-project loading: 88.6%
6. Facility: SHINGLE CRK – SOUTHTOWN 138 KV
Contingency: SOUTHTOWN – VIENNA OPERATIONS 345 KV
Loading: 128.9% (Rating: 250 MVA)

DFAX: 54.3%
Pre-project loading: 91.7%

7. Facility: SPENCER – DOBBINS SUBSTATION 138 KV

Contingency: CADY – DODD ROAD 345 KV
Loading: 124.7% (Rating: 265 MVA)
DFAX: 17.0%
Pre-project loading: 113.8%

8. Facility: EASTON – DOBBINS 138 KV line

Contingency: CADY – DODD RD 345 KV
Loading: 124.7% (Rating: 251 MVA)
DFAX: 17.0%
Pre-project loading: 113.1%

9. Facility: EASTON – NAVY MOLE 138 KV

Contingency: DODD RD – CADY 345 KV L
Loading: 124.2% (Rating: 225 MVA)
DFAX: 17.0%
Pre-project loading: 111.3%

10. Facility: MENOMONIE – WATTSVILLE SUBSTATION 138 KV

Contingency: TASLEY 345/138kV Transformer
Loading: 122.7% (Rating: 214 MVA)
DFAX: 100.0%
Pre-project loading: 42.8%

11. Facility: MESQUITE – ARROYO TAP 138 KV

Contingency: SOUTHTOWN SUBSTATION – VENNA OPERATIONS 345 KV line
Loading: 119.2% (Rating: 220 MVA)
DFAX: 45.6%
Pre-project loading: 83.8%

12. Facility: CANDELARIA ARENAS 1718 – OCEAN CITY 138 KV

Contingency: VIENNA OPERATIONS – SOUTHTOWN 345 KV L
Loading: 118.4% (Rating: 197 MVA)
DFAX: 45.6%
Pre-project loading: 78.8%

13. Facility: ARROYO TAP – REIDLAND 138 KV

Contingency: SOUTHTOWN – VIENNA OPERATIONS 345 KV
Loading: 116.9% (Rating: 210 MVA)
DFAX: 45.6%
Pre-project loading: 79.8%

14. Facility: MENOMONIE – COMMONWEALTH CHESAPEAKE 138 KV

Contingency: TASLEY SUBSTATION 345/138 kV XFMR
Loading: 116.7% (Rating: 239 MVA)
DFAX: 100.0%
Pre-project loading: 45.3%

15. Facility: RIPPLE – MESQUITE 138 KV

Contingency: SOUTHTOWN – VIENNA OPERATIONS 345 KV
Loading: 109.1% (Rating: 268 MVA)
DFAX: 45.6%
Pre-project loading: 80.0%

16. Facility: SADDLEBROOKE RANCH – SUNSET 138 KV

Contingency: VIENNA OPERATIONS SUBSTATION – EASTON 2 SUBSTATION 345 KV
Loading: 108.0% (Rating: 413 MVA)
DFAX: 42.0%

Pre-project loading: 90.6%

17. Facility: HANSFORD – SOONER 138 KV

Contingency: SNSET – SADDLEBROOKE RANCH 138 KV line

Loading: 107.8% (Rating: 204 MVA)

DFAX: 10.0%

Pre-project loading: 99.4%

18. Facility: SADDLEBROOKE RANCH – LAMMERS 138 KV

Contingency: EASTON 2 – VIENNA OPERATIONS 345 KV

Loading: 103.5% (Rating: 632 MVA)

DFAX: 47.7%

Pre-project loading: 90.6%

19. Facility: NAVY MOLE – HOLSTON DAM 138 KV

Contingency: EATON 2 – VIENNA OPERATIONS SUBSTATION 345 KV

Loading: 103.4% (Rating: 284 MVA)

DFAX: 9.7%

Pre-project loading: 97.5%

20. Facility: HANSFOD – SADDLEBROOKE RANCH 138 KV

Contingency: SADDLEBROOKE RANCH – SUNSET 138 KV

Loading: 101.7% (Rating: 250 MVA)

DFAX: 10.0%

Pre-project loading: 94.8%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
WATTSILLE – CHINCOTEAGUE 138 KV line	Rebuild line with higher-capacity conductor	\$6,804,000	100% (\$6,804,000)
W -STATION – ST BONIFACIUS 138 KV Line	Rebuild line with higher-capacity conductor	\$6,421,000	85% (\$5,463,000)
W -STATION – SHINGLE CRK 138 KV Line	Rebuild line with higher-capacity conductor	\$12,552,000	85% (\$10,680,000)
SUNSET – ST BONIFACIUS 138 KV	Rebuild line with higher-capacity conductor	\$7,227,000	85% (\$6,149,000)
REIDLAND – OCEAN CITY SUBSTATION 138 KV Line	Rebuild line with higher-capacity conductor	\$7,459,000	83% (\$6,180,000)
SHINGLE CRK – SOUTHTOWN 138 KV	Rebuild line with higher-capacity conductor	\$4,297,000	85% (\$3,656,000)
SPENCER – DOBBINS SUBSTATION 138 KV	Rebuild line with higher-capacity conductor	\$4,862,000	100% (\$4,862,000)
EASTON – DBBINS 138 KV	Rebuild line with higher-capacity conductor	\$1,948,000	100% (\$1,948,000)
EASTON – NAVY MOLE 138 KV	Rebuild line with higher-capacity conductor	\$7,076,000	100% (\$7,076,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
MENOMONIE – WATTSVILLE SUBSTATION 138 KV	Rebuild line with higher-capacity conductor	\$6,403,000	100% (\$6,403,000)
MESQUITE – ARROYO TAP 138 KV	Rebuild line with higher-capacity conductor	\$3,874,000	83% (\$3,210,000)
CANDELARIA ARENAS 1718 – OCEAN CITY 138 KV	Rebuild line with higher-capacity conductor	\$3,547,000	83% (\$2,939,000)
ARROYO TAP – REIDLAND 138 KV	Rebuild line with higher-capacity conductor	\$1,301,000	83% (\$1,078,000)
MENOMONIE – COMMONWEALTH CHESAPEAKE 138 KV	Rebuild line with higher-capacity conductor	\$9,356,000	100% (\$9,356,000)
RIPPLE – MESQUITE 138 KV	Reconductor line	\$1,416,000	83% (\$1,173,000)
SADDLEBROOKE RANCH – SUNSET 138 KV	Reconductor line	\$1,894,000	68% (\$1,294,000)
HANSFORD – SOONER 138 KV	Reconductor line	\$1,744,000	16% (\$281,000)
SADDLEBROOKE RANCH – LAMMERS 138 KV	Reconductor line	\$2,453,000	51% (\$1,258,000)
NAVY MOLE – HOLSTON DAM 138 KV	Reconductor line	\$754,000	100% (\$754,000)
HANSFOD – SADDLEBROOKE RANCH SUBSTATION 138 KV	Reconductor line	\$2,739,000	16% (\$442,000)
	Total allocated to project		\$81,006,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.